哈尔滨工业大学深圳校区

综合论文开题报告

题　　目	强化学习驱动的闭环航拍图像识别方法
姓　　名	
学　　号	2023112149
学　　院	
专　　业	
指 导 教 师	
日　　期	2025.12.23

目　录

1 课题背景及研究的目的和意义 3
1.1 3
1.2 3
2 国内外研究现状及分析 4
2.1 4
2.1.1 Visual Grounding 4
2.1.2 4
2.1.3 5
2.1.4 CoT 5
2.2 6
2.2.1 6
2.2.2 7
3 主要研究内容及研究方案 8
3.1 8
3.1.1 8
3.1.2 MLLM 8
3.1.3 9
3.2 10
3.2.1 RL 10
3.2.2 CoT 10
3.2.3 Function Call 10
3.2.4 11
3.2.5 12
3.2.6 13
4 进度安排及预期目标 15
4.1 15
4.2 15
5 已具备和所需的条件和经费 17
5.1 17
5.2 17
6 预计困难及解决方案 18
6.1 18

6.2 18
参考文献 19

1 课题背景及研究的目的和意义

1.1 课题背景

[1][2]

1.2 研究的目的和意义

2 国内外研究现状及分析

2.1 国内外研究现状

2.1.1通用视觉定位（Visual Grounding）的发展演进

Visual Grounding, VG

-

Two-stage Faster R-CNN

RNN/LSTM

Open-vocabulary

Transformer [3][4] One-stage

TransVG, MDETR

2.1.2多模态大模型在视觉定位中的应用，

MLLM GPT-4V LLaVA

Qwen-VL

Reasoning-based VG

Token Kosmos-2 [4]

Location as Language

GRIT Grounded Image-Text

Token Location Tokens

Kosmos-2

Referring Expression Comprehension Referring

Expression Generation

Qwen-VL[8] Visual

Receptor

Qwen-VL 448x448

Position-aware Adapter

Image-Caption-Box

Tuples Zero-shot

Qwen2.5-VL[7]

Native Dynamic Resolution

Qwen2.5-VL MROPE Multimodal Rotary Position Embedding

Fine-grained Visual Perception

2.1.3无人机航拍场景下的视觉感知挑战

MLLM UAV

Domain Gap

COCO[6]

4K/8K

0.1% MLLM

336x336 448x448

Viewpoint Variation

2.1.4强化学习与思维链（CoT）在视觉推理中的应用

RL

Static Inference

Vision Reasoner[10], Ground-R1[12]

RL

- DeepSeek-R1[5]

Chain-of-Thought, CoT Seg-Zero[11] UniVG[2]

RL

RL MLLM

2.2 国内外文献综述及简介

2.2.1现有研究的局限性

RefCOCO/+/g6

MSCOCO

RefCOCO+

RefCOCOg

VisDrone UAVDT

Hallucination

DMLLMs[4][7][8]

[9]

One-pass Inference

2.2.2强化学习策略研究现状

RL MLLMs

VG Vision Reasoner[10] RL

Seg-Zero[11]

RL GRPO [5] CoT

UniVG-R1[2] CoT GRPO

SOTA

Ground-R1 CoT RL

RL

MLLMs VG

COT

GRPO

3 主要研究内容及研究方案

3.1 研究内容

LLM

COT

3.1.1引入强化学习策略

Markov Decision Process, MDP

Agent state

action

3.1.2使用多模态大语言模型（MLLM）作为智能体

MLLM

agent MLLM

Chain of Thought, CoT

1. 2. 3. 4.

3-1

3.1.3闭环系统

MLLM

LLM COT

- -

3.2 研究方案

Observe

Think Act Refine

3.2.1构建基于强化学习（RL）的主动视觉感知策略网络。

MDP Policy Network

State Action

IoU PPO

GRPO

3.2.2搭建具备思维链（CoT）能力的多模态推理基座。

Qwen2-VL

LLaVA-Next

SFT

Chain-of-Thought, CoT

->

-> -> ->

3.2.3实现基于 Function Call 的闭环系统与自我修正机制。

Function Call

-

CLIP Score

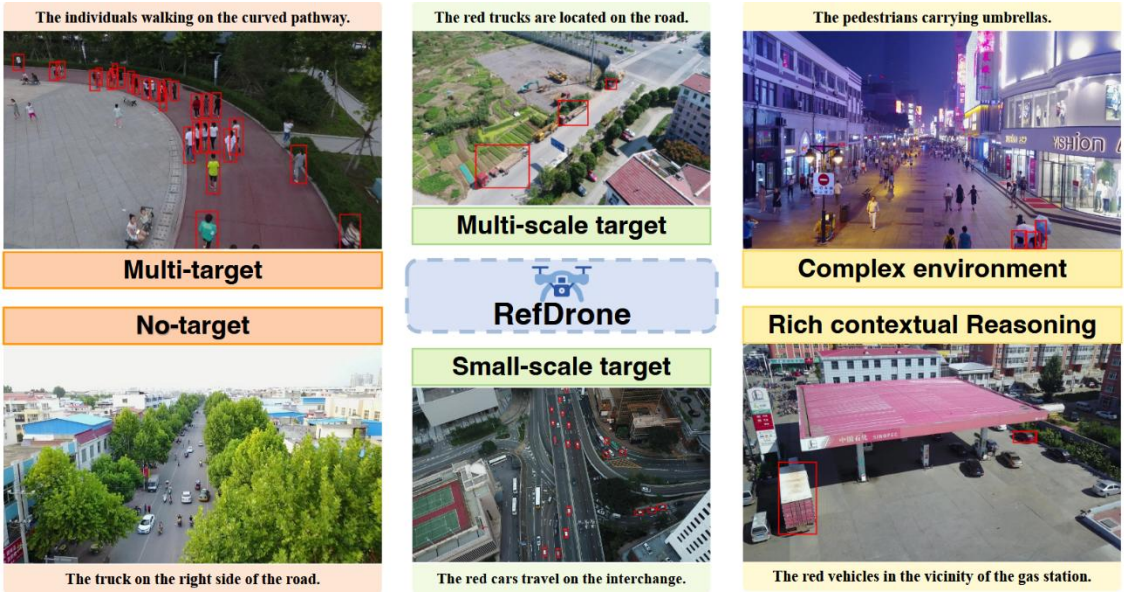
Reflection RL

- - -

3.2.4测试数据集

+

3-1



3-2 RefDrone

3-1

VisDrone-DET 2021	2021	10,209 260 +	60% <0.1%	**RL Agent**
RefDrone	2025	8,536 17,900	/ 31% 14%	

3-1

AerialVG	2024	5,000	**CoT**
		50,000	SFT
		( )	CoT

3.2.5评价指标

(Accuracy@IoU) (mAP)
(Reasoning Consistency)

(1) (Intersection over Union, IoU)

B_{pred} B_{gt} IoU

$$IoU = \frac{area(B_{pred} \cap B_{gt})}{area(B_{pred} \cup B_{gt})}$$

(2) (Accuracy@k) $IoU \geq k$ $k = 0.5$ N

:

$$Acc@k = \frac{1}{N} I \sum_{i=1}^{N} (IoU_i \geq k)$$

I(·) 1 0

$Acc@0.5$ $Acc@0.75$

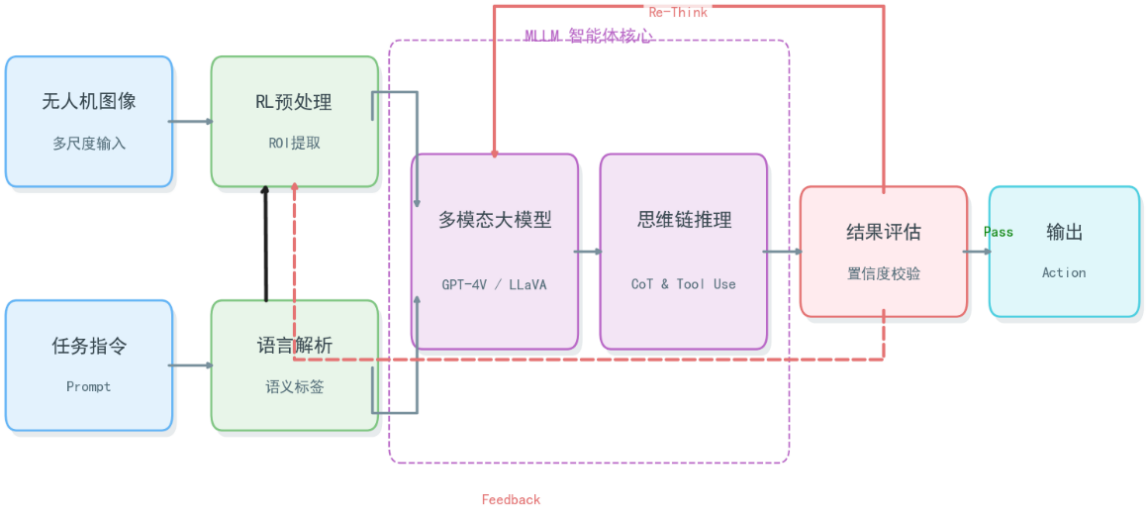
(3) (Reward Function)

$Agent$ R_t

$$R_t = \lambda_1 \cdot \left(IoU_t - IoU_{t-1}\right) + \lambda_2 \cdot I\left(IoU_t > 0.5\right) - \lambda_3 \cdot Cost_{step}$$

3.2.6实施方案与技术路线。

-



3-3

VisDrone

GPT-4o

LoRA

AerialVG

4 进度安排及预期目标

4.1 进度安排

2025 08 -2025 10 DeepSeek GRPO Qwen-VL AerialVG VisDrone

-

-

2025 11 -2026 01 Transformer LLM Agent

2026 02 -2026 03 LLM Agent Reward Engineering

- -

2026 03 -2026 04 CoT SOTA

2026 04 -2026 05 PPT Demo

4.2 预期目标

MLLM RL

-

Chain-of-Thought，CoT

AerialVG RefDrone SOTA

mAP

IoU@0.5

Function Call

5 已具备和所需的条件和经费

5.1 实验室条件和经费保障

GPU

5.2 所需条件和经费

Google Colab AutoDL

GPT-4o Claude-3.5 API CoT Distillation API

6 预计困难及解决方案

6.1 预计困难与技术难点

VG

6.2 解决方案

CoT Distillation

INT4/INT8

Curriculum Learning RL

Agent

LoRA Low-Rank Adaptation QLoRA 4-bit/8-bit

Early Stopping

SAM - Segment Anything Model -

参考文献

[1] LIU J, et al. AerialVG: A challenging benchmark for aerial visual grounding by exploring positional relations[R/OL]. arXiv preprint arXiv:2504.07836, 2025.

[2] BAI S, et al. UniVG-R1: Reasoning guided universal visual grounding with reinforcement learning[R/OL]. arXiv preprint arXiv:2505.14231, 2025.

[3] LIU S, ZENG Z, REN T, et al. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection[C]// Proceedings of the European Conference on Computer Vision (ECCV). Cham: Springer, 2024: 38-55.

[4] PENG Z, WANG W, DONG L, et al. Kosmos-2: Grounding multimodal large language models to the world[R/OL]. arXiv preprint arXiv:2306.14824, 2023.

[5] SHAO Z, et al. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models[R/OL]. arXiv preprint arXiv:2402.03300, 2024.

[6] YU L, POIRSON P, YANG S, et al. Modeling context in referring expressions[C]// Proceedings of the European Conference on Computer Vision (ECCV). Cham: Springer, 2016: 69-85.

[7] BAI S, et al. Qwen2.5-VL technical report[R/OL]. arXiv preprint arXiv:2502.13923, 2025.

[8] BAI J, BAI S, YANG S, et al. Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond[C]// Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2024: 1-10.

[9] XIAO L, YANG X, LAN X, et al. Towards visual grounding: A survey[J]. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025: 1-20.

[10] LIU Y, et al. VisionReasoner: Unified visual perception and reasoning via reinforcement learning[R/OL]. arXiv preprint arXiv:2505.12081, 2025.

[11] LIU Y, et al. Seg-Zero: Reasoning-chain guided segmentation via cognitive reinforcement[R/OL]. arXiv preprint arXiv:2503.06520, 2025.

[12] CAO M, ZHAO H, ZHANG C, et al. Ground-R1: Incentivizing grounded visual reasoning via reinforcement learning[R/OL]. arXiv preprint arXiv:2505.20272, 2025.

[13] REDMON J, DIVVALA S, GIRSHICK R, et al. You only look once: Unified,

real-time object detection[C]// Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2016: 779-788.

[14] SHI Y, DI S, CHEN Q, et al. Enhancing video-LLM reasoning via agent-of-thoughts distillation[C]// Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2025: 8523-8533.

[15] YANG J, ZHANG H, CHEN X, et al. LLM-Grounder: Open-vocabulary 3D visual grounding with large language model as an agent[C]// Proceedings of the IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2024: 7694-7701.

[16] SUN Z, et al. RefDrone: A challenging benchmark for referring expression comprehension in drone scenes[R/OL]. arXiv preprint arXiv:2502.00392, 2025.